

AVIATION LAW AND POLICY SERIES

KLUWER LAW INTERNATIONAL

Aviation Safety through the Rule of Law

ICAO's Mechanisms and Practices

JIEFANG HUANG



Wolters Kluwer
Law & Business

KLUWER INTERNATIONAL



30806081

Aviation Safety through the Rule of Law

ICAO's Mechanisms and Practices

Jiefang Huang



Wolters Kluwer

Law & Business

AUSTIN BOSTON CHICAGO NEW YORK THE NETHERLANDS

**Published by:*

Kluwer Law International
PO Box 316
2400 AH Alphen aan den Rijn
The Netherlands
Website: www.kluwerlaw.com

Sold and distributed in North, Central and South America by:

Aspen Publishers, Inc.
7201 McKinney Circle
Frederick, MD 21704
United States of America
Email: customer.service@aspublishers.com

Sold and distributed in all other countries by:

Turpin Distribution Services Ltd.
Stratton Business Park
Pegasus Drive, Biggleswade
Bedfordshire SG18 8TQ
United Kingdom
Email: kluwerlaw@turpin-distribution.com

Printed on acid-free paper.

ISBN 978-90-411-3115-7

© 2009 Kluwer Law International BV, The Netherlands.

Graphic Design: Flora Huang

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without written permission from the publisher.

Permission to use this content must be obtained from the copyright owner. Please apply to: Permissions Department, Wolters Kluwer Legal, 76 Ninth Avenue, 7th Floor, New York, NY 10011-5201, USA. Email: permissions@kluwerlaw.com

Printed in Great Britain.

Volume 5

Series Editor

Pablo Mendes de Leon

International Institute of Air and Space Law, Leiden University

The global liberalization regime of the past two decades has fundamentally affected the operation of international air transport services.

This process calls for reflection and analysis across a wide range of legal and policy areas, including but not limited to public international air law and policy, with special reference to safety, security and environmental concerns, liability, competition law regimes, company law, and the complex relationship between European Community law and public international law.

The *Aviation Law and Policy Series* critically examines developments around liberalization of air services internationally resulting in the establishment of Open Aviation Areas; institutional questions such as the external powers of the European Community and the relationship between international organizations such as ICAO and EUROCONTROL; the coming into force of unlimited liability regimes and related case law; the granting of antitrust immunity to international airline alliances; and cooperation between competition authorities. Attention will also be paid to topical safety and security issues, and the growing impact of aviation in relation to the environment, dictating the design of emission trade systems. Last but not least, commercially oriented subjects such as aircraft financing and leasing will be addressed, both from the perspective of national and international legal regimes, taking into account practical cases and case law.

The objective of the *series* is to make a contribution to legal thinking on the multifaceted aspects of this important field of law. The publications are designed to document and anticipate the evolution of aviation law and policy in the twenty-first century.

The titles published in this series are listed in the back of this volume.

About the Author

Jiefang Huang is Professor of Law at Civil Aviation University of China (CAUC). He graduated from Wuhan University, China, obtained an LL.M. from the Institute of Air and Space Law, McGill University, Canada, and PhD from the International Institute of Air and Space Law, Leiden University.

He has worked as a student counsellor in Wuhan University and as Attaché/Third Secretary in the Embassy of the People's Republic of China to Canada. He worked from 1988 to 1996 in Bloomfield, Bellemare, a law firm in Montreal representing several member States of the International Civil Aviation Organization (ICAO). In 1996, he joined ICAO Secretariat as a legal adviser.

He has been working as Legal Officer in the Legal Affairs and External Relations Bureau, ICAO. He is also the Dean of the Law School of China Civil Aviation University; concurrent professor in the Institute of International Law, Wuhan University, China; and Vice-President of the International Law Association (Canadian Branch).

To my wife and my parents

Foreword

In all modes of traffic and transport, safety is a primordial issue. This is the case too, and especially so, for civil aviation and air transport. This book deals with aviation safety. As long ago as 1944, the nations of the world agreed upon an international aviation safety regime in the Convention on International Civil Aviation, which, amongst other things, created the International Civil Aviation Organization (ICAO), headquartered in Montreal, and empowered it to issue safety standards and recommended practices, and to see to their oversight.

The author of this book, Dr Jiefang Huang, wrote his doctoral thesis at Leiden University on the subject. The book is a commercial version of the dissertation, earlier published, in a limited edition, by the E.M. Meijers Institute for Legal Research at Leiden's Law Faculty. The book addresses both aviation safety and the functions and practices of ICAO exhaustively. It also devotes attention to aviation security. Whereas, in ordinary parlance, safety and security are synonymous, in aviation the word security has taken on the special meaning of safeguarding aviation from unlawful interference, of which terrorism is the most dangerous form.

Dr Huang's most significant contribution to legal science, it seems, is his plea to consider civil aviation safety obligations not only as merely contractual obligations between States, but as obligations owed *erga omnes*, to the international community as a whole, and as generally applicable, compulsory law for all (*jus cogens*). This plea will no doubt give rise to interesting discussions and follow-up in the years to come.

The book, as the product of a Ph.D. thesis, is not only doctrinally, theoretically thorough, but it also profits greatly from the practical experience of the author as a legal officer at ICAO. It has been my pleasure and privilege to have been Dr Huang's thesis supervisor (*promotor*) at Leiden University from 2001 to

2009, and equally to have been his teacher, much earlier, in the mid-1980s, when he was a graduate student for a master's (LL.M.) degree at McGill University.

Dr P.P.C. Haanappel
Professor of Air and Space Law (em.), Leiden University
Adjunct Professor of Law, McGill University
March 2009

Acknowledgement

I would like to express my most profound gratitude to Professor Peter P.C. Haanappel, Professor of Air and Space Law at the International Institute of Air and Space Law, Leiden University, who has guided my study in air law twice in my life at both McGill University and Leiden University. Without his enthusiastic motivation, invaluable supervision, and kind assistance, it would not have been possible to complete this work.

My deep appreciation is conveyed to Mr Denys Wibaux, Director of the Legal Affairs and External Relations Bureau of ICAO, for sharing his vast experience in international law and for his unfailing support of my work; to Mr John Augustin, Senior Legal Officer of ICAO, for inspiration given through his thesis and for reading some of my manuscript; and to other colleagues in ICAO for a constant exchange of ideas. Of course, the views expressed in the thesis are those of the writer, and I myself shall be solely responsible for all errors and weaknesses that may be found in it.

Special indebtedness is owed to Dr Niels van Antwerpen, corporate legal counsel for the legal department of KLM Royal Dutch Airlines, for sharing his previous experience in Leiden's PhD programmes, for translating certain material, and for rendering every possible assistance in the process of my study, including acting as my paramin in the final defence of my thesis.

My sincere thanks also go to Ms Marla Weinstein and Ms Arlene Tyo of ICAO as well as Mr Yaw Nyampong of McGill University for proofreading my manuscript and providing editorial assistance in the English language; to Ms Tatiana Koukharskaia of the ICAO Legal Affairs and External Relations Bureau for rechecking the entire manuscript; to Ms Lydia Nawfal of the same bureau for formatting a part of the electronic version of the manuscript; to Ms Paula van der Wulp, Office Manager of the International Institute of Air and Space Law of Leiden University for all administrative assistance; and to Ms Karin van

Heijningen of the E.M. Meijers Institute and Ms Anne-Marie Krens for assisting the publication.

After the submission of the manuscript to the Meijer Institute for publication of a limited edition, I had the privileges to receive valuable comments from, in chronological order, Professor Pablo Mandes de Leon, Director of the International Institute of Air and Space Law, Leiden University; Professor Bin Cheng, Former Dean of the Law Faculty, London University; Mr Gilbert Guillaume, Former President, International Court of Justice; Professor Neri Sybesma-Knol, Vrije Universiteit Brussels; Professor Nico Schrijver, Academic Director of the Grotius Centre for International Legal Studies, Leiden University; Madame Folasade Oduola, Director, ICAO Air Transport Bureau; Mr Mohamed Elamiri, Chief, ICAO Safety and Security Audit Branch; Mr Armando Quiroz, Chief, ICAO Security Audit Section; and Mr Silvério Espinola, ICAO Principal Legal Officer. My gratefulness is registered for every one of them. In particular, I wish to note that Professor Bin Cheng, despite his senior age and his retirement for many years, has written seven pages of comments. Some of his comments have been reflected here, other insightful remarks will still need further digestion in my future endeavours.

I could not forget to mention the loyal support of my wife, LEI Min (雷敏), who has unilaterally shouldered the entire family obligations and burden over the years to allow me to concentrate on my work. As a small token of my appreciation, I managed to complete my thesis exactly by her birthday, 4 November 2008. I also remember the love, support, and encouragement given by my parents, Mr HUANG Ing Hui (黄永晖) and Madame CHAN Ching Mui (陈静梅), and by my brother, WONG Wai Hung (黄卫红). To Flora and Lucia, I would apologize for the temporary absence of fatherhood.

I wish to register my thanks to the Government of the Kingdom of the Netherlands for making the educational system in Leiden University available to me. I also record my lifetime gratitude to the Government of the People's Republic of China for providing me with a scholarship to study abroad, when foreign currencies were still scarce resources in the country.

Jiefang Huang
Tianjin, China,
2 April 2009

List of Abbreviations and Acronyms

AASL	Annals of Air and Space Law
ACAC	Arabic Civil Aviation Commission
AFCAC	African Civil Aviation Commission
AJIL	American Journal of International Law
ANC	Air Navigation Commission
ASECNA	L'Agence pour la sécurité de la navigation aérienne en Afrique et à Madagascar
ATM	Air Traffic Management
BYIL	British Yearbook of International Law
C-Dec	ICAO Council Decision
Chicago Convention	Convention on International Civil Aviation
Chinese JIL	Chinese Journal of International Law
C-Min.	ICAO Council Minutes
CNS/ATM	Communications, Navigation, Surveillance/Air Traffic Management
COCESNA	Central American Corporation for Air Navigation Services
CYIL	Canadian Yearbook of International Law
DGCA	Directors General of Civil Aviation
EASA	European Aviation Safety Agency
EC	European Community
ECAC	European Civil Aviation Conference
EJIL	European Journal of International Law
EU	European Union
EUROCONTROL	European Organisation for the Safety of Air Navigation
FAA	Federal Aviation Administration of the United States

FANS	Future Air Navigation Systems
FIR	Flight Information Region
FUA	Flexible Use of Airspace
GNSS	Global Navigation Satellite System
ICAN	International Commission for the Air Navigation
ICAO	International Civil Aviation Organization
ICJ	International Court of Justice
ICTY	International Criminal Tribunal for the Former Yugoslavia
ILC	International Law Commission
ILM	International Legal Materials
JAA	Joint Aviation Authority
JALC	Journal of Air Law and Commerce
LACAC	Latin American Aviation Commission
LJIL	Leiden Journal of International Law
LNTS	League of Nations Treaty Series
MEX Convention	Convention on the Marking of Plastic Explosives for the Purpose of Detection
Montreal Convention	Convention for the Suppression of Unlawful Acts against the Safety of Civil Aviation
Montreal Supplementary Protocol	Protocol for the Suppression of Unlawful Acts of Violence at Airports Serving International Civil Aviation, Supplementary to the Convention for the Suppression of Unlawful Acts against the Safety of Civil Aviation, done at Montreal on 23 September 1971
OJ	Official Journal of the European Union
PICAO	Provisional International Civil Aviation Organization
PKD	Public Key Directory
RdC	Recueil des cours de l'Académie de droit international de la Haye (Collected Courses of the Hague Academy of International Law)
SAFA	Safety Assessment of Foreign Aircraft
SARPs	International Standards and Recommended Practices
SUPPS	Regional Supplementary Procedures
The Hague Convention	Convention for the Suppression of Unlawful Acts against the Safety of Civil Aviation
Tokyo Convention	Convention on Offences and Certain Other Acts Committed on Board Aircraft
UN	United Nations
UNGA	United Nations General Assembly
UNTS	United Nations Treaty Series
US	United States of America

USAP	Universal Security Audit Programme
USOAP	Universal Safety Oversight Audit Programme
ZLW	Zeitschrift für luft und Weltraumrecht

Table of Contents

About the Author	vii
Foreword	xi
Acknowledgement	xiii
List of Abbreviations and Acronyms	xix
Introduction	1
Chapter 1	
Defining Aviation Safety in View of the Global Interest	5
1.1 The Concept of Aviation Safety	6
1.2 Historical Development of Aviation Safety Regulations	8
1.3 Renewed Importance of Aviation Safety in Contemporary Society	15
1.4 Aviation Safety: The Raison D'être of ICAO	17
1.5 Concluding Remarks	21
Chapter 2	
Regulation of Aviation Safety by Means of a Technical Safety Code	23
2.1 Establishment of Safety Oversight Responsibility	23
2.1.1 Responsibility of the State of Registry and/or the State of the Operator	24
2.1.2 Responsibility of States in Their Respective Territories	34
2.1.3 Critical Elements of the Safety Oversight System	42
2.2 Formulation of Technical Regulations	44
2.2.1 Subject Matters Addressed by SARPs	45
2.2.1.1 Personnel Licensing (Annex 1)	46
2.2.1.2 Rules of the Air (Annex 2)	47

2.2.1.3	Airworthiness of Aircraft (Annex 8)	48
2.2.1.4	Operation of Aircraft (Annex 6)	49
2.2.1.5	Other Annexes	49
2.2.2	Criteria for Safety Regulations: Uniformity, Reliability, and Affordability	50
2.2.3	Processes for Formulating Technical Regulations	54
2.2.4	Juridical Nature of Technical Regulations	58
2.2.5	Scope of Application of Technical Regulations	65
2.3	Auditing of State Compliance with Technical Regulations	68
2.3.1	Universal Safety Oversight Audit Programme	69
2.3.2	Universal Security Audit Programme	71
2.3.3	Legal Issues Arising from ICAO Audits	72
2.3.3.1	The Principle of Consent and the Mandatory Nature of Audits	73
2.3.3.2	Confidentiality and Transparency	76
2.3.3.3	Implications for International Law and Practice	77
2.4	Concluding Remarks	81
Chapter 3		
Protecting Aviation Safety from Military Operations		83
3.1	Prohibition of the Use of Weapons against Civil Aircraft in Flight	83
3.1.1	Article 3 <i>bis</i> and Customary International Law	84
3.1.2	Reactions of ICAO to the Use of Weapons against Civil Aircraft	89
3.1.3	Freedom of Action in Times of War or National Emergency	92
3.1.4	Revisiting Article 3 <i>bis</i> in the Context of 11 September 2001	94
3.2	Coordination of Potentially Hazardous Activities to Civil Aircraft	100
3.3	General Relations between Civil and Military Aviation	104
3.4	Concluding Remarks	109
Chapter 4		
Strengthening Aviation Safety against Unlawful Interference		111
4.1	The Tokyo Convention	112
4.2	The Hague Convention	119
4.3	The Montreal Convention	126
4.4	The Montreal Supplementary Protocol	133
4.5	The MEX Convention	139
4.6	Addressing New and Emerging Threats after 11 September 2001	147
4.7	Characterization of Crimes against the Safety of Civil Aviation	150
4.8	Concluding Remarks	155

Chapter 5

Enhancing Aviation Safety through the Rule of Law	157
5.1 Safety Obligations and Fundamental Norms	157
5.1.1 Safety and Obligations <i>Erga Omnes</i>	157
5.1.1.1 Obligations towards 'the International Community as a Whole'	159
5.1.1.2 'The Importance of the Rights Involved'	161
5.1.1.3 Concern of All States	163
5.1.1.4 Positive Prescriptions of Obligations <i>Erga Omnes</i>	164
5.1.2 Safety Obligations and <i>Jus Cogens</i>	167
5.2 Safety and ICAO'S Regulatory Function	174
5.2.1 The Legal Effect of ICAO Assembly Resolutions	175
5.2.1.1 Declaratory Resolutions	176
5.2.1.2 Interpretative Resolutions	178
5.2.1.3 Pre-legislative Resolutions	178
5.2.1.4 Directive Resolutions	181
5.2.1.5 Recommendatory Resolutions	183
5.2.2 Concept of Quasi-Legislation or Quasi-Law	187
5.2.2.1 Doctrinal and Constitutional Basis for Quasi-Law	187
5.2.2.2 Essence of Quasi-Law	192
5.2.2.3 Scope of Quasi-Law	194
5.2.2.4 Effect of Quasi-Law	195
5.2.2.5 Implementation of Quasi-Law	197
5.3 ICAO Enforcement and Implementation Functions	199
5.3.1 Enforcement Functions	200
5.3.1.1 Enforcement under Article 54(j) of the Chicago Convention	200
5.3.1.2 Other Enforcement Mechanisms	202
5.3.2 Implementation Functions	203
5.4 Implementation at the National and Regional Level	204
5.4.1 Implementation in National Law in General	205
5.4.1.1 Monism	205
5.4.1.2 Dualism	206
5.4.1.3 Sui Generis Approach	207
5.4.2 Regional Implementation	207
5.4.3 Global Ramifications of National and Regional Initiatives	210
5.4.4 Reverted Monism?	214
5.5 Checks and Balances under the Rule of Law	217
5.5.1 Checks and Balances in ICAO Quasi-Judicial Activities	222
5.5.2 Checks and Balances of ICAO Quasi-Legislative Activities	222
5.6 Concluding Remarks	226
General Conclusions	229
Selected Bibliography	237
Index	253

Introduction

In November 1944, towards the end of the Second World War, representatives from fifty-four nations gathered in Chicago to design a blueprint for the worldwide regulation of post-war international civil aviation.¹ The conference resulted in the adoption of the Convention on International Civil Aviation (Chicago Convention) on 7 December 1944 and the establishment of the International Civil Aviation Organization (ICAO) on 4 April 1947, when the Convention came into force.² The main mission of ICAO is to 'insure the safe and orderly growth of international civil aviation throughout the world'.³ Accordingly, since the date of its birth, ICAO has been closely linked with aviation safety.⁴

More than sixty years have passed. Has ICAO lived up to the expectations of its founders? Some believe that it is 'one of the most effective international organizations in the United Nations system';⁵ others, while praising the work of its first

-
1. P.P.C. Haanappel, *The Law and Policy of Air Space and Outer Space: A Comparative Approach* (The Hague: Kluwer Law International, 2003), 17. Fifty-two States signed the Final Act of the Chicago Conference. See, The Department of State, United States of America, *Proceedings of the International Civil Aviation Conference*, vol.1 (Washington: United States Government Printing Office, 1948). See also, *infra* Ch. 1, section 1.2.
 2. See, ICAO Doc. 7300/9 *Convention on International Civil Aviation*.
 3. Article 44 a), Chicago Convention. The Preamble also mentions that the undersigned governments have agreed on certain principles and arrangements in order that international civil aviation 'may be developed in a safe and orderly manner'.
 4. See, *infra* Ch.1, sections 1.2 to 1.4. Throughout the present study, the term 'safety' or 'aviation safety' refers to the safety of international civil aviation. It does not deal with the use of aircraft for military services, or the law of air warfare, except to the extent that they have impact on the safety of international civil aviation.
 5. A.J. Broderick & J. Loos, 'Government Aviation Safety Oversight – Trust, but Verify', *JALC* 67 (2002): 1035–1036.